

CERONIX SUPPLY CHAIN RISK ANALYSIS SYSTEM

Comprehensive Architecture Documentation

With Detailed Diagrams and Technical Specifications

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EXECUTIVE SUMMARY

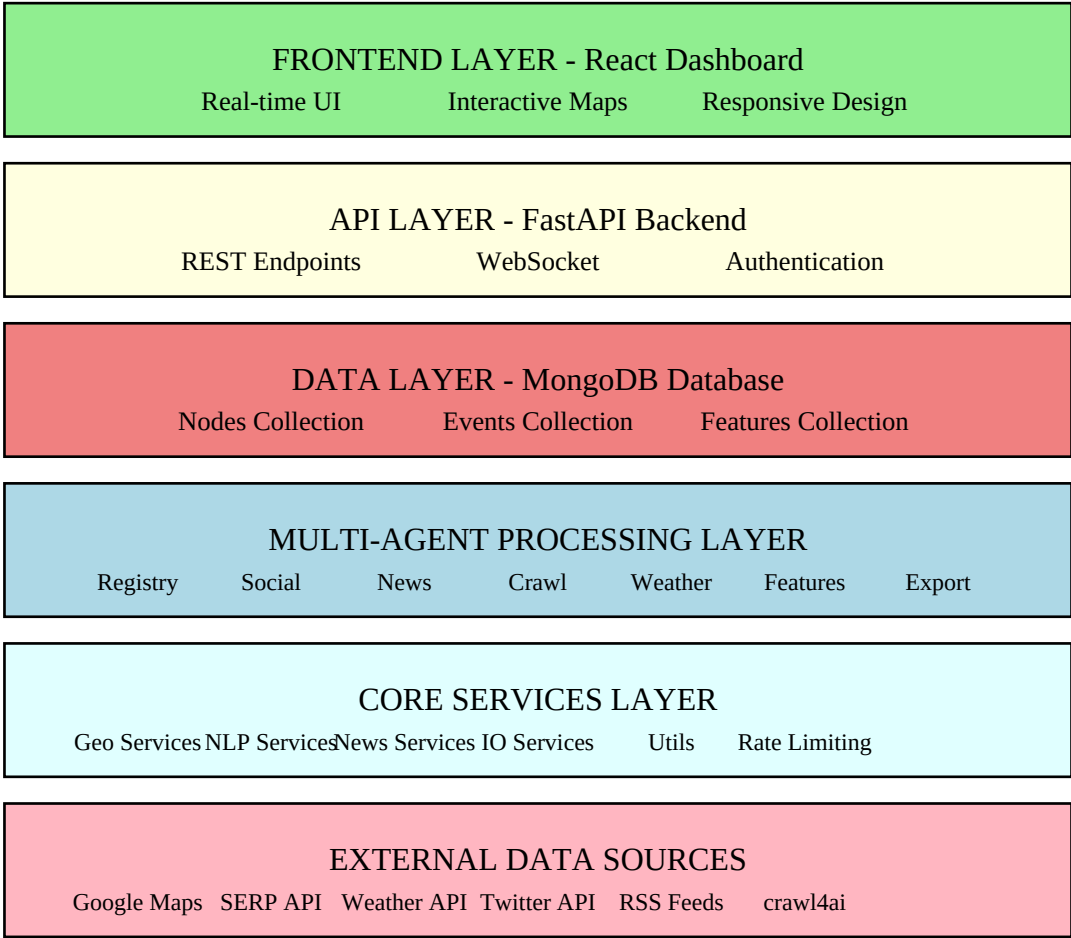
The CERONIX Supply Chain Risk Analysis System is a comprehensive, production-ready platform designed to provide real-time visibility and analysis of supply chain risks. The system combines advanced multi-agent architecture with modern web technologies to deliver actionable insights through an intuitive dashboard interface.

Key Benefits:

- Real-time risk monitoring and alerting
- Automated data collection from multiple sources
- Intelligent feature engineering for ML models
- Scalable architecture supporting growth
- Interactive visualization and reporting
- Production-ready deployment with Docker

DETAILED SYSTEM ARCHITECTURE

CERONIX SUPPLY CHAIN RISK ANALYSIS SYSTEM



The system follows a layered architecture pattern with six distinct layers:

Frontend Layer: React-based dashboard providing real-time visualization, interactive maps, and responsive design for optimal user experience across devices.

API Layer: FastAPI backend offering RESTful endpoints, WebSocket support for real-time updates, and comprehensive authentication mechanisms.

Data Layer: MongoDB database with optimized collections for nodes, events, features, and analytics, supporting horizontal scaling.

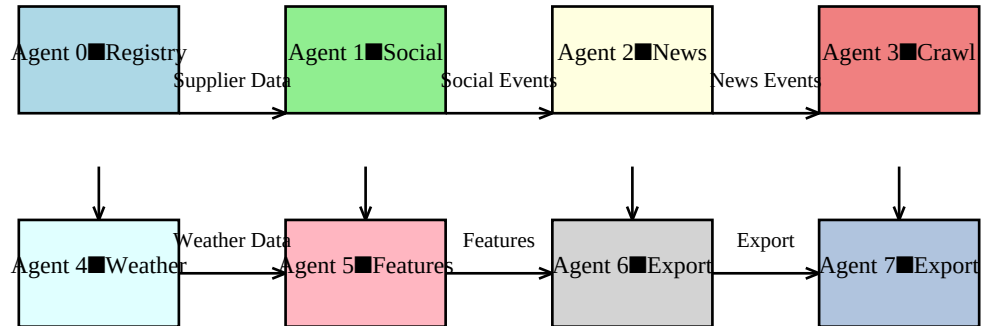
Multi-Agent Processing: Eight specialized agents working in coordination to collect, process, and analyze data from various sources.

Core Services: Reusable service modules providing geocoding, NLP, news processing, and utility functions across the system.

External Data Sources: Integration with Google Maps, SERP API, Weather API, Twitter API, and RSS feeds for comprehensive data collection.

AGENT INTERACTION FLOW

AGENT INTERACTION FLOW

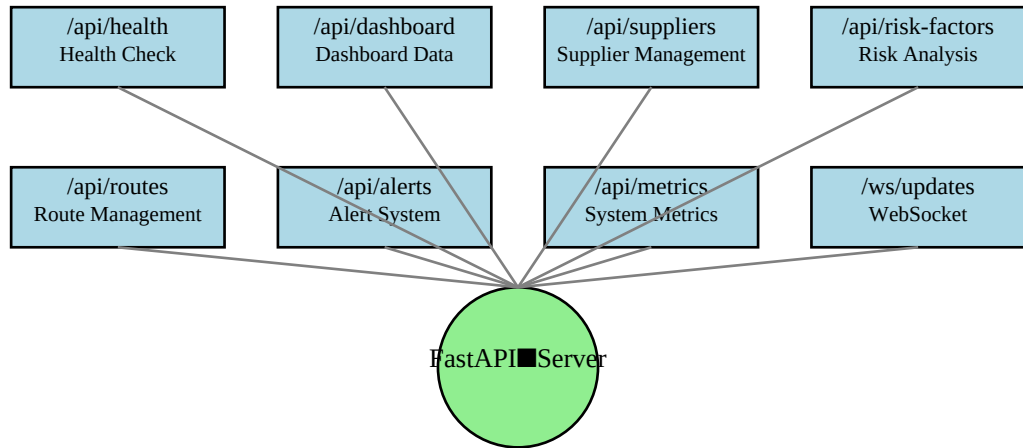


The multi-agent system operates through coordinated data flow:

1. Agent 0 (Registry) processes seed supplier data and performs geocoding
2. Agents 1-3 collect social media, news, and web content data
3. Agent 4 monitors weather conditions and detects anomalies
4. Agent 5 performs feature engineering and combines all data sources
5. Agents 6-7 handle export operations and generate reports
6. All agents store processed data in MongoDB for API consumption

API ENDPOINT ARCHITECTURE

API ENDPOINT ARCHITECTURE



The REST API provides comprehensive endpoints for system interaction:

Health Monitoring: `/api/health` - System status and component health checks

Dashboard Data: `/api/dashboard` - Aggregated data for dashboard visualization

Supplier Management: `/api/suppliers` - CRUD operations for supplier data

Risk Analysis: `/api/risk-factors` - Risk metrics and analysis results

Route Management: `/api/routes` - Supply chain route information

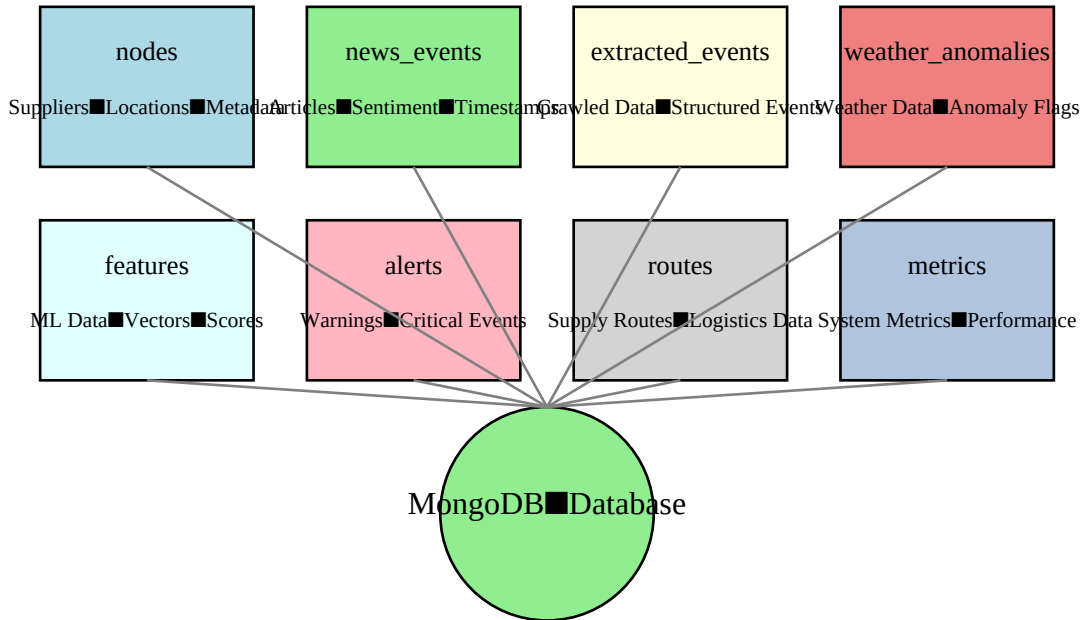
Alert System: `/api/alerts` - Real-time alerts and notifications

System Metrics: `/api/metrics` - Performance and business metrics

Real-time Updates: `/ws/updates` - WebSocket for live data streaming

DATABASE SCHEMA ARCHITECTURE

MONGODB DATABASE SCHEMA



MongoDB collections are optimized for performance and scalability:

nodes: Stores supplier information, locations, and metadata with geospatial indexing

news_events: Contains news articles with sentiment analysis and relevance scoring

extracted_events: Holds structured events extracted from web crawling operations

weather_anomalies: Weather data with anomaly detection flags and risk scores

features: ML-ready feature vectors and computed risk metrics

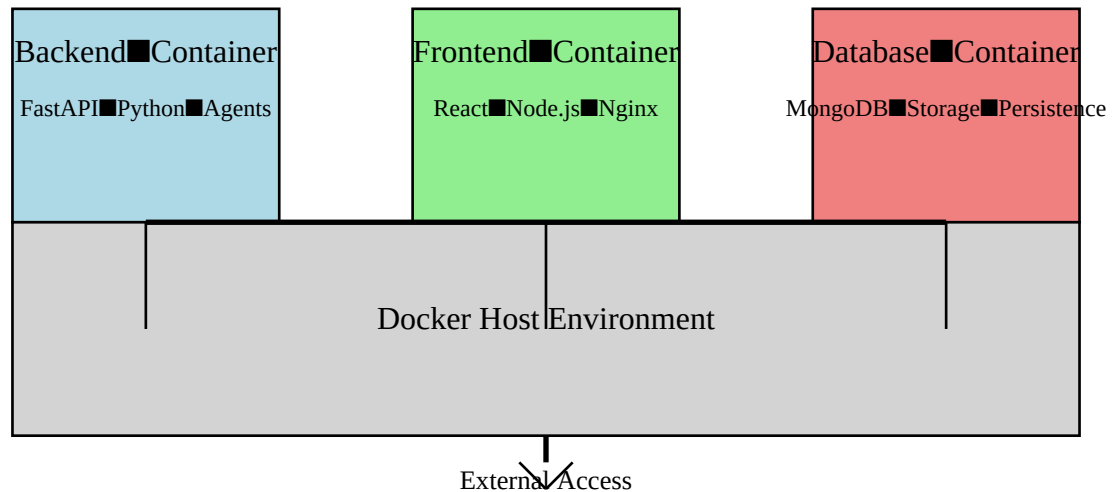
alerts: System alerts, warnings, and critical event notifications

routes: Supply chain route information and logistics data

metrics: System performance metrics and business intelligence data

DEPLOYMENT ARCHITECTURE

DEPLOYMENT ARCHITECTURE



The system is containerized using Docker for easy deployment:

Backend Container: Runs FastAPI server, Python agents, and core services with optimized resource allocation

Frontend Container: Serves React application with Nginx for static file serving and load balancing

Database Container: MongoDB instance with persistent storage and automated backup capabilities

Docker Host: Production environment with monitoring, logging, and security configurations

TECHNICAL SPECIFICATIONS

Component	Technology	Version	Purpose
Frontend	React	18.x	User Interface
Frontend	JavaScript	ES6+	Client Logic
Frontend	CSS3	Latest	Styling
Backend	FastAPI	0.100+	API Server
Backend	Python	3.11+	Core Language
Backend	Pydantic	2.x	Data Validation
Database	MongoDB	6.0+	Data Storage
Database	Motor	3.x	Async Driver
Orchestration	Prefect	2.0+	Workflow Management
Deployment	Docker	Latest	Containerization
Deployment	Nginx	Latest	Web Server

PERFORMANCE CHARACTERISTICS

- Asynchronous Processing: Full async/await implementation for optimal concurrency
- Rate Limiting: Configurable throttling per API endpoint to respect external service limits
- Caching Strategy: Multi-level caching with in-memory and file-based storage
- Horizontal Scaling: Architecture supports load balancing and service replication
- Real-time Updates: WebSocket implementation for live data streaming
- Database Optimization: Indexed queries and connection pooling for performance

SECURITY ARCHITECTURE

Environment Security: API keys and secrets managed through environment variables with proper encryption

API Security: CORS configuration, rate limiting, input validation, and authentication mechanisms

Data Security: MongoDB encryption at rest, secure network connections, and data anonymization

Infrastructure Security: Docker security best practices, network isolation, and access controls

MONITORING & LOGGING

Structured Logging: JSON-formatted logs with timestamps, context, and correlation IDs

Performance Metrics: System performance, business metrics, and user activity tracking

Health Monitoring: Component health checks, service availability, and automated alerting

Error Tracking: Comprehensive error logging with stack traces and context information

FUTURE ROADMAP

- Machine Learning Integration: Direct ML model deployment and prediction serving
- Advanced Analytics: Graph-based risk propagation and network analysis
- Multi-language Support: Internationalization for global supplier networks
- Mobile Application: Native mobile apps for on-the-go monitoring
- Advanced Caching: Redis integration for distributed caching
- Microservices Architecture: Service decomposition for better scalability
- Real-time Streaming: Apache Kafka integration for high-volume data processing
- Advanced Security: OAuth 2.0, RBAC, and audit logging implementation

CONCLUSION

The CERONIX Supply Chain Risk Analysis System represents a comprehensive solution for modern supply chain risk management. With its multi-agent architecture, real-time processing capabilities, and intuitive web interface, the system provides organizations with the tools needed to identify, analyze, and mitigate supply chain risks effectively.

The architecture is designed for scalability, maintainability, and performance, ensuring that the system can grow with organizational needs while providing reliable, real-time insights into supply chain operations.